Executive Summary

Customer Engagement & Product Utilization Analytics for Retention Strategy

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1. Project Overview

Customer retention is an important concern for banking institutions because losing existing customers can affect long-term customer relationships and business performance. Understanding customer behavior, engagement, and banking product utilization can help institutions identify retention patterns and develop more informed customer engagement strategies.

The project *Customer Engagement & Product Utilization Analytics for Retention Strategy* examines customer churn using a European banking dataset containing 10,000 customer records. The primary focus is to understand how customer engagement, the number of banking products held, account balance, and credit card ownership relate to customer retention.

The project was developed using Python for data analysis and visualization, with an interactive Streamlit dashboard to present the findings.

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2. Objectives

The main objectives of the project were:

- To examine the relationship between customer engagement and churn.
- To analyse churn rates across different banking product counts.
- To identify inactive customers with relatively high account balances.
- To compare the Relationship Strength Index of retained and churned customers.
- To examine the relationship between credit card ownership and retention.
- To develop an interactive dashboard for exploring customer retention patterns.

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3. Methodology

The analysis followed a quantitative and exploratory approach. The dataset was loaded and examined using Python libraries, including Pandas and NumPy. Data validation identified 10,000 customer records, no missing values, no duplicate rows, and 10,000 unique customer IDs.

Exploratory data analysis was performed to understand the distribution of customer engagement, product ownership, account balances, and churn status. Additional analytical variables were created to support customer segmentation and relationship-strength analysis.

A custom Relationship Strength Index was developed by combining engagement and product usage scores. The findings were presented through visualizations and an interactive Streamlit dashboard featuring key performance indicators, customer filters, product utilization analysis, and a high-value disengaged customer module.

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4. Key Findings

The analysis produced several important findings.

Overall customer churn: Of the 10,000 customers, 2,037 had churned and 7,963 were retained. The overall churn rate was 20.37%.

Customer engagement: Inactive customers had a churn rate of 26.85%, compared with 14.27% among active customers. This difference indicates an association between customer inactivity and higher churn in the dataset.

Product utilization: Customers holding two banking products had the lowest observed churn rate, at 7.58%. Customers holding one product had a churn rate of 27.71%, while the rates for customers holding three and four products were 82.71% and 100%, respectively. The latter groups were relatively small, so their results require cautious interpretation.

High-balance customers: Customers in the high-balance group had a churn rate of 23.68%, compared with 19.27% among lower-balance customers. Furthermore, 49.88% of high-balance customers were classified as inactive.

Relationship Strength Index: Retained customers had a mean Relationship Strength Index of 47.03, compared with 36.48 among churned customers. This suggests that the custom index differs between the two customer groups, although further validation is needed.

Credit card ownership: The retention rate was 79.82% among customers with credit cards and 79.19% among customers without them. The observed difference was relatively small.

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5. Business Implications and Recommendations

The findings suggest that customer retention analysis can benefit from considering engagement, product utilization, and financial characteristics together.

Banks could monitor engagement indicators to identify inactive customer groups for further investigation. Customers with high account balances but low engagement could be reviewed to determine whether additional communication or service support would be appropriate.

Product bundling decisions should consider actual customer needs and utilization rather than assuming that holding more products automatically improves retention. Similarly, credit card ownership should be considered alongside other customer characteristics rather than treated as a standalone measure of customer loyalty.

The custom Relationship Strength Index may provide an additional exploratory indicator for customer segmentation. However, it should be tested and validated before being used to guide operational retention decisions.

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6. Dashboard Development

An interactive Streamlit dashboard was developed to communicate the analytical findings. It presents customer summary indicators, engagement-related churn patterns, product utilization comparisons, high-value disengaged customer information, and Relationship Strength Index analysis.

The dashboard also provides filters for geography, engagement status, number of products, account balance, and estimated salary. These features allow users to explore customer groups and examine retention patterns interactively.

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7. Conclusion

The project demonstrates how Python-based data analytics and interactive visualization can support the study of banking customer retention.

The findings highlight a notable association between customer engagement and churn, differences in retention across product-count groups, and higher observed churn among high-balance customers compared with lower-balance customers. The Relationship Strength Index provides an additional exploratory approach to examining customer relationships.

Overall, the project emphasizes the value of evaluating customer behavior and product utilization alongside financial characteristics. The results can support further investigation into customer retention strategies, although additional statistical testing and validation would be required before the findings could be used for operational predictions or decisions.

Project Outcome: An exploratory customer retention analytics solution consisting of data analysis, customer segmentation, key performance indicators, visualizations, and an interactive Streamlit dashboard.
